VisAI Labs

Brainchild of e-con Systems

VisAI 2023-03

Instructions:

- Candidates are asked to fill in the required fields on the sheet attached to the answer sheet.
- This question booklet carries 6 subjective Puzzle questions & 20 multiple choice questions in C language.
- Do not write anything on the question paper.
- There is no negative marking.
- All answers to be written & marked in the answer sheet.
- Rough sheets to be used for solving the problems and to be tied with the Answer sheet.
- The duration for completing this question paper is: 1Hr. 30 min (45 mins + 45 mins).

Section A - Puzzles

Duration: 45 minutes

1. There are three people (Alex, Ben and Cody), one of whom is a knight, one a knave, and one a spy. The knight always tells the truth, the knave always lies, and the spy can either lie or tell the truth. Alex says: "Cody is a knave." Ben says: "Alex is a knight." Cody says: "I am the spy." Who is the spy?

2. In a party attended by married couples, no husband or wife shakes each other's hands but everyone else shakes hands exactly once. There were 180 handshakes. How many couples were at the party?

3. A man has 10 ties in his drawer: 3 blue, 3 black and 4 red. The lights are out, and he is completely in the dark. What is the probability that the first red tie he took was in the third attempt?

4. Prem and his best girl decided to take a bus ride, but on account of his limited resources it was necessary that they should walk back. Now, if the bus goes at the rate of nine miles an hour and they walk at the rate of three miles an hour, how far can they ride so that they may be back in eight hours?

5. In a village there are 100 people, from 1900 to 1999 each person has been born. In 1999, If a person's age is equal to sum of his birth year. What is his birth year?

6. There are 20 pieces of bread to be divided among 20 people. A man eats 3 pieces, woman eats 2 pieces, and a child eats half piece of bread. Tell the correct combination of men, women and children so that they are 20 people in total, and everyone gets the bread. Note that a man cannot eat less than 3 or more than 3. A woman cannot eat less than 2 or more than 2 and the child cannot eat less than half or more than half a piece of the bread. How many women is in the group?

Section B – Programming MCQs

Duration: 45 minutes

7.
```
#include<stdio.h>
int main()
{
    char ch;
    ch = 'A';
    printf("The letter is");
    printf("%c\n", ch >= 'A' && ch <= 'Z' ? ch + 'a' - 'A':ch);   'a',
    printf("Now the letter is");
    printf("%c\n", ch >= 'A' && ch <= 'Z' ? ch : ch + 'a' - 'A');
    return 0;
}
```

A. The letter is a
Now the letter is A
B. The letter is A
Now the letter is a
C. Error
D. None of above

8.
```
#include<stdio.h>
int get();

int main()
{
   const int x = get(); 20
   printf("%d", x);
   return 0;
}
int get()
{
   return 20;
}
```

A. Garbage value
B. Error
C. 20
D. 0

9.
```
#include <stdio.h>
void show();
int main()
{
   #ifndef CVV
      #define CVV 199
   #endif
   show();
   return 0;
}
void show()
{
   printf("CVV=%d",CVV);
}
```

CVV

CVV

A. No output
B. CVV=0
C. CVV=199
D. Compiler error

10. What is the output of C program?

```
#include<stdio.h>
int main()
{
   float a=654.123456f;
   printf("%3.3f,%3.2f",a,a);
   return 0;
}
```

A. 654,654
B. 654.123456,654.123456
C. 654.123,654.12
D. 654.000,654.00

11. #include<stdio.h>

```
void myshow();
int main()
{
        myshow();
        myshow();
        myshow();
}
void myshow()
{
        static int k = 20;
        printf("%d ", k);
        k++;
}
```

A. 20 20 20
B. 20 21 21
C. 20 21 22
D. Compiler error.

12. #include<stdio.h>

```
# include <stdio.h>
void fun(int *ptr)
{
   *ptr = 30;
}
int main()
{
 int y = 20;
 fun(&y);
 printf("%d", y);
 return 0;
}
```

A. 20
B. 30
C. Compiler Error
D. Runtime Error

13.
```
#include <stdio.h>
int main()
{
    int counter = 1;
    do {
        printf("%d, ", counter);
        counter += 1;
    } while (counter >= 10);
    printf("\nAfter loop counter=%d", counter);
    printf("\n");
    return 0;
}
```

A. After loop counter=1

B. 1,
After loop counter=2

C. 1,
After loop counter=1

D. After loop counter=2

14.
```
#include <stdio.h>
#include <stdlib.h>
int main()
{
        int m = 2, c=1;
        int n,a,b, limit=10;
        while(c < limit)
        {
                for (int n = 1; n < m; ++n)
                {
                        a = m * m - n * n;
                        b = 2 * m * n;
                        c = m * m + n * n;
                        if (c > limit)
                        break;
                        printf("%d %d %d\n", a, b, c);
                }
                m++;
```

```
        }
    }
```

A. 5 3 7

B. 2 5 7
7 9 10

C. 1 2 3
4 5 6
7 8 9

D. 3 4 5
8 6 10

15.
```
#include<stdio.h>
    void fun1(char *s1, char *s2) {
        char *tmp;
        tmp = s1;
        s1 = s2;
        s2 = tmp;
    }

    void fun2(char **s1, char **s2) {
        char *tmp;
        tmp = *s1;
        *s1 = *s2;
        *s2 = tmp;
    }

    int main () {
        char *str1 = "VisAI", *str2 = "Labs";
        fun1(str1, str2);
        printf("%s %s ", str1, str2);
        fun2(&str1, &str2);
        printf("%s %s ", str1, str2);
        return 0;
    }
```

A. VisAI Labs Labs VisAI
B. Labs VisAI VisAI Labs
C. Labs ViaAI Labs VisAI
D. VisAI Labs VisAI Labs

16.
```
#include<stdio.h>
int func(int n)
{
   int i=0;
   while(n%10!=0)
   {
      n=n+3;
      i++;
   }
   n=n-i;
   return n;
}
void main()
{
   printf("%d",func(35));
}
```
A. 50
B. 55
C. 53
D. 45

17.
```
#include<stdio.h>
void main()
{
   int a=5,b=2,c=1;
   if(b>a && a>c && c>b)
      b=a+1;
   else
      a=b+1;
   printf("%d",a+b+c);
}
```

A.6
B.11
C.19
D.7

18.
```
#include <stdio.h>
int main()
{
  float a = 5;
  switch(a){
  case 5: printf("econ");
  default: printf("system");
  }
  return 1;
}
```
A.econ
B.system
C.econsystem
D.Error

19.
```
#include<stdio.h>
int main()
{
int i,j,count;
count=0;
for(i=0; i<5; i++);
{
for(j=0;j<5;j++);
{
   count++;
}
}
printf("%d",count);
return 0;
}
```

A. 55
B. 54
C. 1
D. 0

20. What kind of expression is "new Point (2,3)"?
A. Constructor Calling Expression
B. Invocation Expression
C. Object Creation Expression
D. Primary Expression

21. Which one of the given options can be considered as the correct output for the following JavaScript code?

```
const obj1 =
{
a:10,
b:15,
c:18
};
const obj2 =Object.assign({c:7, d:1}, obj1);
console.log(obj2.c, obj2.d);
```

A. Undefined
B. 18,1
C. 7,1
D. Error

22. If the following piece of JavaScript code is executed, will it work if not, what kind of possible error can occur?

```
function fun(o)
{
for(;o.next; oo =o.next);
return o;
}
```

A. Yes, it will work fine
B. No, this will not iterate at all
C. No, it will throw an exception as only numeric's can be used in a for loop

D. No, it will produce a runtime error with the message "Cannot use Linked List"

23. Consider the following program, where are i, j and k are stored in memory?

```
int i;
int main()
{
    int j;
   int *k = (int *) malloc (sizeof(int));
}
```

A. i, j and *k are stored in stack segment
B. i and j are stored in stack segment. *k is stored on heap.
C. i is stored in BSS part of data segment, j is stored in stack segment. *k is stored on heap.
D. j is stored in BSS part of data segment, i is stored in stack segment. *k is stored on heap.

24. Assume sizeof an integer and a pointer is 4 byte. Output?

```
#include <stdio.h>

#define R 10
#define C 20

int main()
{
  int (*p)[R][C];
  printf("%d",  sizeof(*p));
  getchar();
  return 0;
}
```

A. 200
B. 4
C. 800
D. 80

25. What is printed by the following C program?

```
#include <stdio.h>
int f(int x, int *py, int **ppz)
{
  int y, z;
  **ppz += 1;
  z = **ppz;
  *py += 2;
  y = *py;
  x += 3;
  return x + y + z;
}
void main()
{
  int c, *b, **a;
  c = 4;
  b = &c;
  a = &b;
  printf( "%d", f(c,b,a));
  getchar();
}
```

A. 18
B. 19
C. 21
D. 22

26. What is the meaning of using extern before function declaration? For example, following function sum is made extern

```
extern int sum(int x, int y, int z)
{
   return (x + y + z);
}
```

A. Function is made globally available
B. extern means nothing, sum() is same without extern keyword.
C. Function need not to be declared before its use
D. Function is made local to the file.